What DNA Foundation Models Learn Beyond Sequence Composition

Anonymous Author(s)
Affiliation
Address
email

Abstract

A central question in genomic representation learning is whether DNA foundation models (FMs) encode information beyond simple sequence composition, or whether their apparent representational power largely reflects signals already captured by classical compositional statistics. We address this by establishing k-mer frequency vectors as a rigorous biologically grounded baseline and introducing a geometric decomposition that partitions FM embeddings into a component linearly predictable from k-mer frequencies and an orthogonal residual representing genuinely non-compositional information. By evaluating each component separately as a downstream feature space, we directly measure what FMs encode beyond composition and whether that content is beneficial, neutral, or detrimental.

Across 57 genomic classification datasets and 2 gene expression regression tasks, we compare five FMs (NTv3-650M, HyenaDNA, DNABERT-2, Caduceus-Ph, Evo2-1B) against k-mer features ($k \in \{4, 5, 6\}$) under a controlled frozen-embedding protocol with a fixed Random Forest classifier. NTv3 is the only model to consistently outperform k-mers (63.2% of datasets, mean MCC $= 0.562$), with gains concentrated in splice-site and promoter tasks; HyenaDNA, DNABERT-2, and Evo2 are outperformed on 86%, 95%, and 93% of datasets. Wilcoxon signed-rank tests with FDR correction confirm these patterns are systematic and robust to dataset heterogeneity.

Our decomposition reveals that FM advantage originates exclusively from the non-compositional residual, and only in tasks with inherently positional regulatory signals. For HyenaDNA and DNABERT-2, the residual actively degrades classification. The fraction of embedding variance explained by k-mers (R^2) is uncorrelated with downstream quality, demonstrating that the quantity of non-compositional information is not a proxy for its utility. These results show that, in the frozen-embedding regime, current DNA FM representations largely recapitulate sequence composition, and provide a principled diagnostic for predicting which task conditions are likely to benefit from non-compositional structure.

Keywords: DNA foundation models, k-mer features, genomic sequence classification, representation learning, computational biology

1 Introduction

Recent advances in genomic foundation models (FMs) [7] mean that large neural architectures [11] can now learn representations directly from raw DNA sequences. Transformer-based architectures and long-context sequence models, such as the Nucleotide Transformer 3 [8], HyenaDNA [23], DNABERT-2 [32], Evo2 [9], and Caduceus [26], have produced promising results in various genomic prediction tasks, such as identifying regulatory elements [4, 3], predicting variant effects [8, 9], and

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modelling gene expression [4]. However, a fundamental question remains unresolved: *do FMs learn representations that meaningfully extend beyond simple sequence composition?*

Many biological signals in genomic sequences are strongly associated with local nucleotide composition and short sequence motifs. One classical representation that captures this information is the k-mer frequency vector [22], which counts the number of times all sequences of length k occur. K-mer representations are computationally simple and interpretable, and have been used for decades in tasks such as metagenomic classification [30, 20] and regulatory element prediction [18]. However, they are rarely considered competitive baselines when evaluating genomic FMs [2, 19, 31, 14, 13, 21]. Consequently, it is unclear whether the performance improvements often attributed to DNA foundation models stem from genuinely richer sequence representations or merely from the absence of robust compositional baselines. If a large proportion of FM embeddings can be predicted directly from k-mer frequencies, the apparent representational power of these models may largely reflect information already captured by simple compositional statistics.

In this work, we systematically investigate what information DNA foundation models encode beyond sequence composition. We establish k-mer frequency vectors as a strong and biologically grounded baseline and introduce a geometric framework that decomposes FM embeddings into two components: one that is predictable from k-mer composition and a residual component that captures non-compositional information. We evaluate this framework across 57 genomic classification datasets [14] and two gene expression regression tasks [29], comparing five widely used DNA foundation models [8, 23, 32, 9, 26] against k-mer representations under a controlled experimental protocol using frozen embeddings and a fixed downstream classifier (Fig. 1).

We address this question through three contributions:

1. we establish k-mer frequency vectors as a rigorous baseline for evaluating the intrinsic representational content of genomic FMs under a frozen-embedding protocol, showing that four of five models do not encode information systematically beyond sequence composition in this regime, while the strongest model (NTv3-650M) achieves gains only under specific task conditions;
2. we introduce a geometric decomposition of FM embeddings into a k-mer-predictable component and a non-compositional residual, enabling a direct measurement of what FMs encode beyond sequence composition; and
3. we show that FM advantage originates exclusively from the non-compositional residual, which is beneficial only for tasks with positional regulatory structure and otherwise neutral or detrimental.

We emphasize that this evaluation addresses representational content, not task-specific performance under fine-tuning. A frozen-embedding protocol is the natural instrument for this question: it isolates what the model has learned to encode during pre-training from what it can learn to predict given additional supervision. Our conclusions therefore characterize the structure of FM representations as feature spaces, and should not be interpreted as upper bounds on achievable performance under end-to-end optimization.

2 Related Work

2.1 DNA Foundation Models and Their Evaluation

The rapid progress of large language models in natural language processing [12, 10] has inspired a new generation of foundation models for genomic sequences. Models such as the Nucleotide Transformer [8], HyenaDNA [23], and DNABERT-2 [32] are pre-trained on large genomic corpora using masked language modelling or next-token prediction objectives, and subsequently used as frozen feature extractors for downstream tasks. More recent work has expanded the design space through scaling and architectural innovation: Evo2 [9] explores genome-scale modelling across diverse organisms; Caduceus [26] introduces equivariant long-range sequence modelling; GROVER [25] demonstrates the capacity of language models to capture sequence context in the human genome. Earlier models, including DNABERT [17] and Enformer [4], established the importance of modelling both local motifs and long-range regulatory interactions. These architectures differ substantially in inductive bias, from transformer encoders with fixed k-mer tokenisation to long-range convolutional

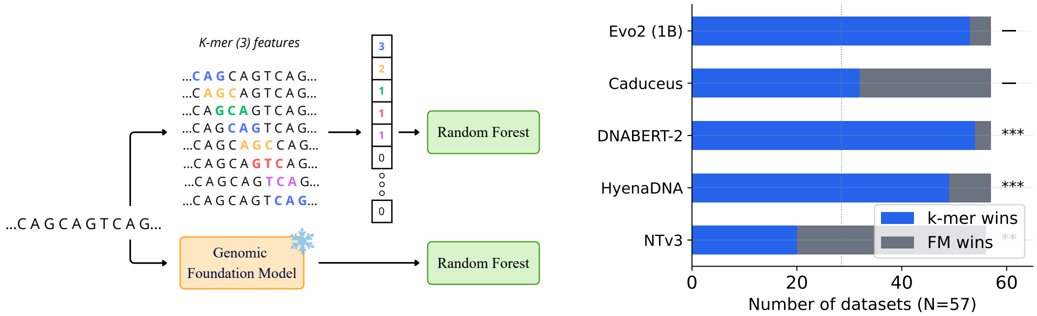

Figure 1: **Left**: experimental setup. Genomic sequences are independently processed into k-mer frequency vectors and FM embeddings, then evaluated using a fixed downstream classifier to directly compare their predictive utility. **Right**: win/loss counts across 57 classification datasets for each FM against the dataset-optimal k-mer baseline. The dashed vertical line marks 28 datasets (50%). NTv3 is the only model to win on the majority of datasets; HyenaDNA, DNABERT-2, and Evo2 are outperformed by k-mers on more than 85% of datasets.

operators with single-nucleotide resolution and subword-based schemes, providing a broad sample of the current design space.

Systematic evaluation of these models has been pursued by several benchmarks [14, 21]. Feng et al. [14] compare multiple models, including DNABERT-2, HyenaDNA, and Nucleotide Transformer variants, using frozen embeddings and standardised downstream classifiers; Marin et al. [21] evaluate DNA language models on biologically meaningful tasks using convolutional prediction heads. Despite providing valuable insights into relative model performance, both benchmarks compare neural architectures primarily against each other and against CNN-based classical baselines, without including simple compositional representations such as k-mer frequency vectors. As a result, they do not resolve whether foundation models outperform biologically grounded but computationally trivial baselines, leaving open the central question of whether FM representations are genuinely richer than sequence composition.

2.2 Compositional Representations and Representation Analysis

k-mer frequency vectors, which encode the distribution of all subsequences of length k, are among the oldest and most widely used representations in computational genomics [18, 30]. They are computationally efficient, interpretable, and directly capture nucleotide composition, which underlies many regulatory signals: transcription factor binding is governed by short sequence motifs, typically 6–12 bp, that are directly expressed as enriched k-mers [27], while CpG dinucleotide density, a key determinant of DNA methylation, is fully specified at the level of 2-mers [6]. Even in modern settings, compositional features remain highly competitive: early deep learning models for regulatory genomics, such as DeepBind [3] and DanQ [24], implicitly learn motif-like representations that closely resemble enriched k-mer patterns, suggesting that much of the predictive signal in genomic sequences is local and compositional. Despite this, k-mer features are rarely treated as a serious baseline in evaluations of DNA foundation models, making it difficult to determine whether improvements attributed to neural architectures reflect genuinely richer representations or simply the absence of a strong compositional reference.

Our work is also related to the probing literature in natural language processing, where auxiliary classifiers are trained on model representations to assess whether specific linguistic properties are linearly decodable [1, 15, 28, 5]. We adopt a complementary geometric perspective: rather than probing for a predefined property, we decompose each FM embedding into a component that is linearly predictable from k-mer frequencies and an orthogonal residual, then evaluate both components as classifier inputs. This approach directly quantifies compositional redundancy and isolates the downstream contribution of non-compositional information, without requiring a task-specific probe.

3 Methods

We compare k-mer frequency vectors and FM embeddings as two symmetric representations under a controlled protocol: both are evaluated on identical datasets using the same fixed downstream classifier and cross-validation scheme, so that performance differences reflect representational quality alone. To move beyond a win/loss comparison, we additionally introduce a geometric decomposition that partitions each FM embedding into a k-mer-predictable component and an orthogonal residual, enabling a direct measurement of what each model encodes beyond local sequence composition.

3.1 Datasets

We evaluate our methods on two complementary dataset collections. For classification, we use 57 binary and multi-class benchmarks from the DNA Foundation Models Benchmark [14], which spans six biological categories: enhancers, promoters, splice sites, transcription factor binding sites, histone modifications, and virus identification.

For regression, we include two gene expression prediction tasks from the Genomics Long-range Benchmark [29]: Bulk RNA Expression and CAGE Prediction. Performance on these tasks is evaluated using R^2. The sequences in these datasets extend up to 6,000 bp, allowing us to move beyond the short-sequence setting and assess whether compositional features and foundation model representations generalize to longer-range prediction tasks.

3.2 Feature Representations

We compare two families of representations, treated symmetrically throughout.

k-mer frequency vectors. For each sequence, we compute the normalized frequency of all subsequences of length k, yielding vectors of dimension 4^k for $k \in \{4, 5, 6\}$. We report results for the dataset-optimal k as the primary baseline, the most favorable condition for k-mers and therefore a conservative reference for FM comparisons.

FM embeddings. We extract mean-pooled frozen embeddings from five DNA foundation models without fine-tuning: NTv3-650M, a transformer encoder pre-trained on 850 genomes using 6-mer tokenization; HyenaDNA-160k, a long-range convolutional model operating at single-nucleotide resolution pre-trained on the human reference genome; DNABERT-2, a BERT encoder with BPE tokenization trained on 135 species; Caduceus-Ph, a bidirectional state-space model based on the Mamba architecture with parallel heads pre-trained on the human reference genome; and Evo2-1B, a large-scale autoregressive model pre-trained across diverse organisms. These five architectures span the main axes of variation in the current design space, scale, tokenization scheme, and sequence modeling approach, making the comparison representative.

3.3 Evaluation Protocol

Both representation families are fed to a Random Forest classifier with 500 trees and default `scikit-learn` hyperparameters, evaluated using stratified 5-fold cross-validation. By fixing the classifier and using frozen embeddings, performance differences primarily reflect representational quality rather than downstream optimization.

We choose Random Forest for two reasons. First, it requires minimal hyperparameter tuning, reducing the risk that observed differences arise from classifier optimization rather than embedding quality. This criterion was independently adopted by Feng et al. [14], who reached the same conclusion. Second, Random Forests naturally handle high-dimensional sparse inputs, which is important given the heterogeneous representation sizes considered here, ranging from 256 dimensions for HyenaDNA to 4,096 for $k = 6$ k-mers, without requiring dimensionality reduction that could differentially affect representations with distinct structural properties.

We report the Matthews Correlation Coefficient (MCC) as the primary classification metric. MCC provides a single scalar summary that remains informative under class imbalance and does not require threshold selection. It is also directly comparable across binary and multi-class tasks; for the latter, we report macro-averaged MCC.

3.4 Geometric Decomposition of FM Embeddings

Formally, let $\mathbf{K} \in \mathbb{R}^{n \times p}$ be the k-mer frequency matrix for n sequences and $\mathbf{X} \in \mathbb{R}^{n \times d}$ the corresponding FM embedding matrix. We fit a Ridge regression $\mathbf{K} \to \mathbf{X}$ on the training fold of each cross-validation split [1], obtaining a weight matrix $\mathbf{W}$. This defines two complementary components:

$$\hat{\mathbf{X}} = \mathbf{K}\mathbf{W} \tag{1}$$

$$\mathbf{E} = \mathbf{X} - \hat{\mathbf{X}} \tag{2}$$

$\hat{\mathbf{X}}$ is the *k-mer-predictable projection*, the part of the FM embedding that lies in the subspace spanned by k-mer features. $\mathbf{E}$ is the *non-compositional residual*, its orthogonal complement, capturing information that cannot be expressed as a linear function of k-mer frequencies. We quantify compositional redundancy via the average coefficient of determination across embedding dimensions:

$$R^2 = \frac{1}{d} \sum_{j=1}^{d} R_j^2 \tag{3}$$

A high R^2 indicates that most embedding variance is linearly predictable from k-mer frequencies. A low R^2 indicates substantial non-compositional content, though notably, not whether that content is biologically useful, a distinction that the downstream evaluation makes explicit.

We evaluate $\hat{\mathbf{X}}$ and $\mathbf{E}$ independently as inputs to the downstream classifier under the same protocol as Section 3.3. Comparing the projection to raw k-mer features tests whether the compositional fraction of the FM adds any discriminative value beyond an explicit frequency count. Comparing the residual to k-mer features isolates the contribution of non-compositional information and determines whether it is beneficial, neutral, or detrimental.

4 Experimental Results

We present results in five parts: classification performance across 57 datasets, regression performance on two gene expression tasks, embedding decomposition by architectural profile, computational efficiency, and the effect of sequence length and GC content on representation performance.

4.1 Classification Performance

Figure 2 plots FM embedding MCC against dataset-optimal k-mer MCC for all 57 datasets across five models. The dominant pattern is that k-mers constitute a strong and largely unbeaten baseline. NTv3 is the only model that consistently outperforms k-mers, winning on 36 of 57 datasets (63.2%, mean MCC 0.562), with win rates reaching 100% on splice-site tasks, 90% on EMP, and above 80% on iPro-WAEL and promoter datasets, dropping to 0% on virus and iDNA-ABF tasks, a pattern consistent with advantage concentrated in tasks with inherently positional regulatory signals. HyenaDNA and DNABERT-2 are outperformed by k-mers on 86.0% and 94.7% of datasets respectively. Evo2 performs worst overall (win rate 6.9%, no wins on splice-site or promoter tasks), likely reflecting a mismatch between its generative pre-training objective and the discriminative frozen-embedding setting. Caduceus wins on 43.9% of datasets but shows a negative mean delta of -0.042, indicating an overall k-mer edge despite occasional FM wins.

Wilcoxon signed-rank tests on the 57 paired Δ values ($\text{MCC}_{\text{FM}} - \text{MCC}_{\text{best k-mer}}$), with Benjamini–Hochberg FDR correction, confirm that all five patterns are systematic (Appendix B, Table 1). The underperformance of HyenaDNA (median $\Delta = -0.074$), DNABERT-2 (-0.095), Caduceus (-0.013), and Evo2 (-0.277) is highly significant (all FDR $q \leq 0.021$) and robust to dataset-level variability under a mixed-effects model. NTv3's positive shift is significant marginally (median $\Delta = +0.024$, FDR $q = 0.007$) but attenuated under the mixed model ($p = 0.081$), confirming that its advantage is task-selective rather than uniform.

4.2 Regression Performance

The regression results (Fig. 3) extend and partially complicate the classification picture. On Bulk RNA expression, NTv3 is the only model to outperform k-mers, achieving $R^2 = 0.46$ compared

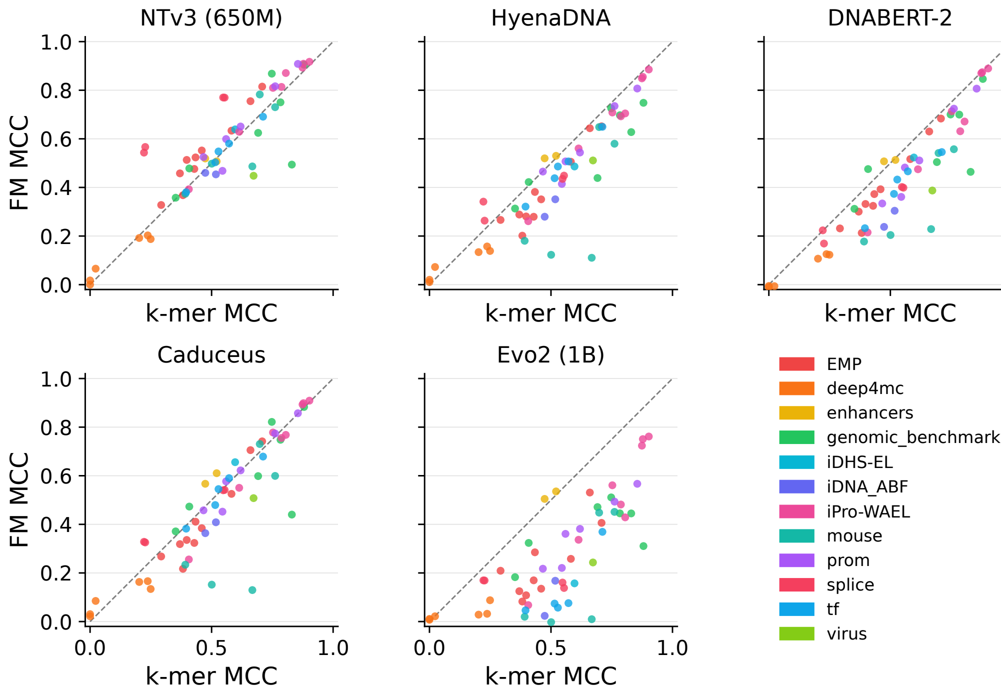

Figure 2: Scatter plots of FM embedding MCC against dataset-optimal k-mer MCC across all 57 classification datasets, for five DNA foundation models. Each point represents one dataset, coloured by biological category. The dashed diagonal indicates equal performance between the FM and k-mer baseline; points above the diagonal indicate FM advantage, points below indicate k-mer advantage. Dataset-optimal k is selected retrospectively across $k \in \{4, 5, 6\}$, representing the most favourable condition for k-mers.

to the k-mer baseline of 0.41, consistent with its advantage on longer sequences observed in the classification setting. HyenaDNA falls just below k-mers ($R^2 = 0.39$), while DNABERT-2 and Caduceus are substantially weaker ($R^2 = 0.19$ and 0.21, respectively).

CAGE prediction reverses this ordering in a striking way: k-mers achieve $R^2 = 0.57$, matching HyenaDNA exactly and outperforming NTv3 ($R^2 = 0.52$), Caduceus ($R^2 = 0.48$), and DNABERT-2 ($R^2 = 0.40$). CAGE peak prediction targets promoter-proximal transcription initiation sites, where the relevant signal is concentrated in local sequence composition around transcription start sites rather than in long-range contextual structure precisely the setting where k-mer features are sufficient and FM embeddings offer no advantage.

Taken together, the regression results confirm that the advantage of foundation models is not a general property of these architectures but depends critically on whether the task requires information beyond local sequence composition.

4.3 Embedding Decomposition

Figure 4 shows the distribution of Ridge R^2 across 57 datasets, quantifying what fraction of each FM embedding is linearly predictable from k-mer frequencies. The models span a wide range: HyenaDNA and Caduceus are the most compositionally redundant (median $R^2 = 0.75$ and 0.71), NTv3 is intermediate ($R^2 = 0.51$), and DNABERT-2 has the lowest redundancy ($R^2 = 0.19$).

The downstream evaluation (Figure 5) reveals a counterintuitive pattern. For DNABERT-2, the projection achieves mean MCC $= 0.510$, above both the full embedding (0.427) and the k-mer baseline (0.535), while the full embedding underperforms the projection in 55 of 57 datasets (mean gap $= -0.083$): the non-compositional residual, which constitutes 81% of the embedding, actively degrades classification. HyenaDNA presents a complementary failure: despite being 75% compositionally redundant, its projection achieves only MCC $= 0.469$ against a k-mer baseline of 0.535,

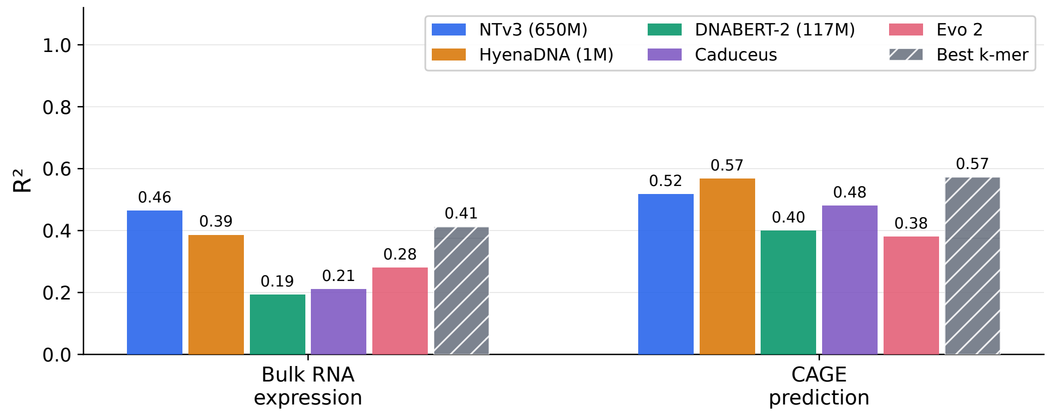

Figure 3: R^2 on two gene expression regression tasks for five DNA foundation models and the dataset-optimal k-mer baseline. On Bulk RNA expression, NTv3 is the only model to exceed the k-mer baseline. On CAGE prediction, k-mers match the best FM (HyenaDNA) and outperform all others, including NTv3.

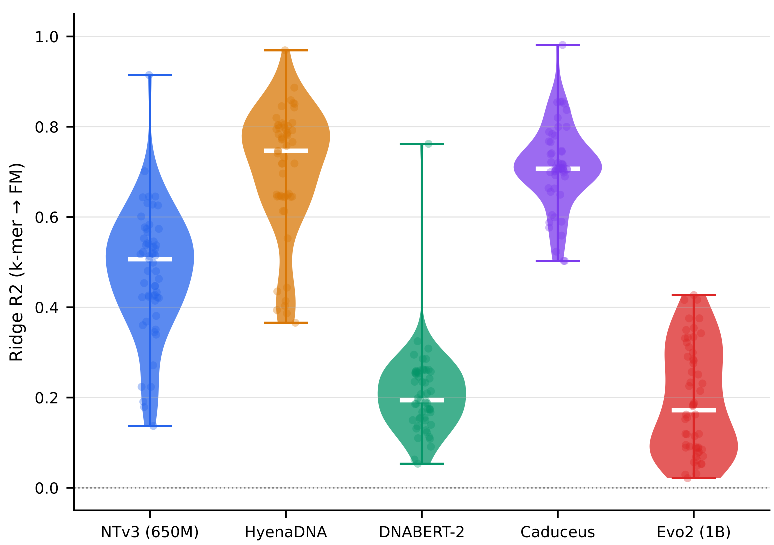

Figure 4: Distribution of Ridge R^2 (k-mer $\rightarrow$ FM embedding, $k = 6$) across 57 classification datasets for each model. The white bar indicates the median; individual points correspond to single datasets. High R^2 indicates that most embedding variance is linearly predictable from k-mer frequencies; low R^2 indicates substantial non-compositional content, though not whether it is biologically useful.

indicating that it encodes compositional information *less* efficiently than a raw frequency count, with the most harmful residual of all four models ($\Delta_{\text{residual}-k\text{-mer}} = -0.233$, k-mer wins in 51 of 57 datasets). Caduceus shares a similar R^2 profile (0.71) but shows a qualitatively different pattern: the full embedding and projection perform almost identically ($\Delta_{\text{full}-\text{proj}} = +0.002$, 29 datasets in each direction), suggesting a neutral rather than harmful residual, though still weaker than k-mers ($\Delta = -0.162$, k-mer wins in 49 of 57 datasets). NTv3 is the only model where the full embedding consistently outperforms its projection: full $>$ proj in 35 of 57 datasets (mean gap $= +0.030$), with residual gains concentrated in splice-site tasks where positional signals cannot be encoded by k-mer frequencies by construction.

These results establish that R^2 is not a useful predictor of FM representational quality: it measures the *quantity* of non-compositional information but not its *quality*. DNABERT-2 is the clearest illustration

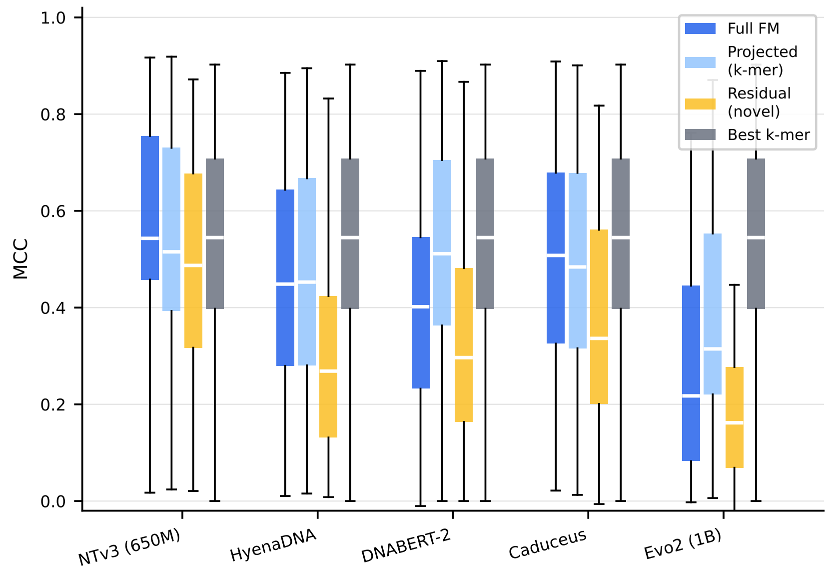

Figure 5: MCC distributions for full FM embedding, k-mer-predictable projection, non-compositional residual, and best k-mer baseline across 57 classification datasets, for five DNA foundation models. Each box spans the interquartile range; the white bar indicates the median. For NTv3, the full embedding marginally but consistently outperforms its projection, the only model where the non-compositional residual adds net value. For HyenaDNA and DNABERT-2, the projection outperforms the full embedding, indicating that the non-compositional residual actively degrades classification. The residual (yellow) falls below the k-mer baseline (grey) for all four models.

that being maximally different from k-mers produces an embedding that is maximally harmful when used in full.

5 Discussion

5.1 k-mer Frequency Vectors as a Biologically Motivated Baseline

The strong performance of k-mer frequency vectors across our benchmark reflects underlying biological structure rather than a fortuitous empirical coincidence. Many regulatory signals in the tasks considered here are inherently compositional: transcription factor binding is governed by short sequence motifs, typically spanning 6–12 base pairs, that are directly captured by k-mer frequencies, while CpG dinucleotide density, a key determinant of DNA methylation, is fully specified at the level of 2-mers [27, 6]. In these settings, local sequence composition provides a sufficient statistic for prediction, and more complex representations offer limited additional benefit.

A direct methodological implication follows: the omission of k-mer baselines in prior benchmarks [14, 21] obscures whether FM performance gains reflect genuinely richer representations or simply the absence of a strong compositional reference. k-mer frequency vectors should be included as a standard baseline in any evaluation of genomic foundation models.

5.2 When Do FM Representations Justify Their Computational Overhead?

Our results define clear conditions under which each model is likely to justify its computational cost over k-mers. NTv3 is the only model for which FM embeddings are reliably beneficial: its advantage is concentrated in tasks with inherently positional regulatory signals, most clearly on splice sites (win rate 100%), EMP (90%), and promoters (83%), and grows with sequence length. For sequences shorter than approximately 300 bp and for tasks such as methylation, TFBS, and enhancers, k-mers match or exceed NTv3 performance at a computational cost that is approximately $10^5\times$ lower. Caduceus wins on 43.9% of datasets but with a negative mean delta relative to k-mers

(-0.042), making it difficult to recommend over k-mers except in specific task settings that remain to be characterised. HyenaDNA, DNABERT-2, and Evo2, do not justify their computational overhead under the frozen-embedding protocol evaluated here: k-mers outperform all three on the large majority of datasets at negligible computational cost.

5.3 What Does Decomposition Reveal About FM Representations?

The geometric decomposition reveals four qualitatively distinct architectural profiles. DNABERT-2's BPE tokeniser generates subword tokens misaligned with biological sequence boundaries, producing an embedding maximally non-redundant with k-mers ($R^2 = 0.19$) whose non-compositional component nonetheless actively degrades classification in 55 of 57 datasets (mean gap $= -0.083$). HyenaDNA exhibits a complementary failure: despite 75% compositional redundancy, its projection achieves only MCC $= 0.469$ against a k-mer baseline of 0.535, indicating that it encodes compositional information *less* efficiently than a raw frequency count. Caduceus shares a similar redundancy profile ($R^2 = 0.71$) but with a neutral residual (mean gap $= +0.002$), suggesting it captures non-compositional structure that is not exploitable under frozen-embedding evaluation. NTv3 is the only model whose residual adds genuine predictive value (full $>$ proj in 35 of 57 datasets, mean gap $= +0.030$), with the largest gains at splice sites where positional signals are inherently non-compositional, consistent with its scale (650M parameters), multi-species pre-training (850 genomes), and 6-mer tokenisation preserving motif boundaries. These profiles confirm that R^2 captures the *quantity* of non-compositional information but not its *quality*: representational utility requires inductive biases aligned with the biological structure of the target task.

5.4 Limitations

Several limitations bound the scope of our conclusions. First, and by design, we evaluate only frozen representations with a fixed Random Forest classifier, the methodologically necessary choice to isolate intrinsic representational content from task-specific optimization capacity. Whether fine-tuning or parameter-efficient adaptation [16] recovers performance on tasks where frozen embeddings underperform k-mers is an orthogonal question that our framework is not designed to answer. Second, a linear probe would more directly test whether the projection/residual distinction holds under a classifier that respects the linear structure of the decomposition. Third, our Ridge-based decomposition is linear and may underestimate the true compositional fraction of FM embeddings. Fourth, all 57 classification datasets derive from a single collection, which may introduce selection bias. Finally, we do not evaluate multi-species generalisation.

6 Conclusion

We have shown that, as fixed feature spaces evaluated without task-specific adaptation, k-mer frequency vectors provide a strong and often sufficient representation for genomic sequence prediction, and that the intrinsic representational content of current DNA foundation models surpasses this baseline only under limited and well-defined conditions. Four of five FM embeddings are largely reducible to sequence composition across 57 classification datasets and two regression tasks, with meaningful gains arising exclusively from a non-compositional residual that captures positional or long-range structure.

The geometric decomposition we introduced provides a practical diagnostic for this distinction, identifying four qualitatively distinct architectural profiles – harmful residual (DNABERT-2), inefficient composition (HyenaDNA), neutral residual (Caduceus), and genuinely informative residual (NTv3) – and demonstrating that R^2 alone is not a reliable guide to representational quality.

These findings characterize what FM pre-training learns to encode, not what FM-based systems can achieve under fine-tuning, a distinction we regard as scientifically meaningful in its own right. Pre-trained genomic models should be understood as conditionally useful: their advantage over k-mers emerges only when the task requires information beyond local sequence statistics and the model's inductive biases are well aligned with that structure. Progress will therefore depend less on scaling alone and more on tokenisation that respects motif boundaries, pre-training objectives that reward positional discrimination, and evaluation protocols that include k-mer frequency vectors as a standard compositional reference.

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Appendix A Model Architectures

We describe the five foundation models evaluated in this work, focusing on the architectural and pre-training characteristics most relevant to interpreting the decomposition results in Section 4.3.

NTv3-650M [8]. A hybrid U-Net/Transformer encoder with 650M parameters, pre-trained on a multi-species corpus of approximately 9 trillion base pairs spanning 24 eukaryotic species using a masked diffusion language modelling objective. Tokenisation uses a sliding 6-mer window, producing $\sim L/6$ tokens for a sequence of length L and preserving biologically meaningful motif boundaries. We extract mean-pooled embeddings over all non-padding tokens ($d = 1536$). The maximum context window is 1 million base pairs.

HyenaDNA [23]. A decoder-based model using the Hyena operator, a data-controlled long convolution, in place of self-attention, enabling sub-quadratic scaling with sequence length. With 1.6M parameters, it operates at single-nucleotide resolution using character-level tokenisation (A/T/C/G), and is pre-trained exclusively on the human reference genome (GRCh38, 3.2 Gbp) with a next-token prediction objective. We use the HyenaDNA-160k variant for classification tasks ($d = 256$, context window 160,000 bp) and HyenaDNA-1M for regression tasks ($d = 256$, context window 1,048,576 bp). Mean pooling is applied over all sequence positions.

DNABERT-2 [32]. A BERT-architecture encoder with 117M parameters, using Byte Pair Encoding (BPE) tokenisation trained on a multi-species corpus spanning 380 species, and ALiBi positional embeddings for length generalisation. BPE produces variable-length tokens (average ~ 4 bp each) that do not align with fixed biological sequence boundaries, driving a higher token count per sequence than 6-mer models at equivalent lengths and multiplying quadratic attention cost. Pre-training uses a masked language modelling objective. We extract mean-pooled embeddings ($d = 768$).

Caduceus-Ph [26]. A bidirectional, reverse-complement equivariant model based on the Mamba state-space architecture (BiMamba blocks with parallel heads), with approximately 35M parameters. It uses character-level tokenisation with reverse-complement augmentation, and is pre-trained on the human reference genome with a masked language modelling objective. The maximum context window is approximately 131,000 bp. We extract mean-pooled embeddings ($d = 256$).

Evo2-1B [9]. A large-scale autoregressive model based on the StripedHyena 2 architecture, which combines input-dependent convolutions of varying ranges with sparse attention layers. With 1 billion parameters, it is pre-trained on OpenGenome2, a curated corpus of approximately 8.8 trillion nucleotides spanning bacteria, archaea, eukaryotes, and phages, using a next-token prediction objective at single-nucleotide resolution. We extract mean-pooled embeddings from the final layer ($d = 1920$).

Appendix B Statistical Analyses

B.1 FM vs. K-mer: Wilcoxon and Mixed-Effects Results

We report formal statistical tests for the FM vs. best k-mer comparisons across all 57 classification datasets. For each model, Δ is defined as MCC(FM) $-$ MCC(best k-mer), so that $\Delta > 0$ indicates FM advantage and $\Delta < 0$ indicates k-mer advantage.

Wilcoxon signed-rank test. For each model, we apply a two-sided Wilcoxon signed-rank test on the 57 paired Δ values to assess whether the distribution of differences is systematically shifted from zero. p-values are corrected for five simultaneous comparisons using the Benjamini–Hochberg FDR procedure.

Mixed-effects model. To control for the substantial variability in task difficulty across the benchmark, we additionally fit a linear mixed-effects model with model identity as a fixed effect and a random intercept per dataset. This isolates the systematic FM vs. k-mer difference from variation attributable to dataset-level characteristics.

Results are reported in Table 1.

Table 1: Statistical comparison of FM vs. best k-mer MCC across 57 classification datasets. $\Delta =$ MCC(FM) − MCC(best k-mer). Wilcoxon p-values are FDR-corrected across five comparisons using the Benjamini–Hochberg procedure. Mixed-effects estimates share the same Δ point estimate by construction; the p-value reflects significance after absorbing dataset-level variability via random intercepts.

Model	**Mean Δ**	**Median Δ**	**Wins / Losses**	**Wilcoxon p (FDR)**	**Mixed-effects p**
NTv3-650M	$+0.027$	$+0.024$	36 / 20	0.007	0.081
HyenaDNA-160k	-0.087	-0.074	8 / 49	1.90×10^{-8}	1.44×10^{-8}
DNABERT-2-117M	-0.108	-0.095	3 / 54	8.43×10^{-10}	1.35×10^{-12}
Caduceus-Ph	-0.042	-0.013	25 / 32	0.021	0.006
Evo2-1B	-0.266	-0.277	4 / 53	5.39×10^{-10}	1.62×10^{-67}

The underperformance of HyenaDNA, DNABERT-2, Caduceus, and Evo2 is highly significant under both tests, confirming that k-mers are systematically superior for these models across the benchmark. NTv3's positive shift reaches significance in the marginal Wilcoxon test but is attenuated in the mixed model, indicating that its advantage is task-selective rather than uniform, consistent with the category-level analysis in Section 4.1, where NTv3 wins concentrate in splice-site (100%) and promoter (83%) tasks and drop to 0% on virus and iDNA-ABF datasets.

Notably, Caduceus presents a dissociation between the two tests: the Wilcoxon p-value (0.021) is marginal while the mixed-effects estimate is more significant ($p = 0.006$). This reflects the fact that Caduceus losses are more uniformly distributed across datasets, producing a consistent but small negative shift, whereas its wins are concentrated in specific categories, inflating within-dataset variance relative to the mixed-model residual.

B.2 Classifier Choice: Random Forest vs. Linear Probe

Our main evaluation uses a Random Forest (RF) classifier throughout, applied symmetrically to all representations. A natural alternative is a linear probe (LP), which directly tests whether the information in a representation is linearly decodable, a property that aligns with the linear structure of our geometric decomposition. We compare RF and LP across all 57 classification datasets for each model and for the k-mer baseline.

Figure 6 plots LP MCC against RF MCC for each method. Points above the diagonal indicate LP advantage; points below indicate RF advantage.

The results reveal a heterogeneous picture across models (Table 2). For NTv3, the two classifiers perform comparably: LP outperforms RF in 25 of 57 datasets and RF outperforms LP in 14, with a non-significant mean difference of $+0.009$ MCC ($p = 0.094$). For k-mer features, the comparison is essentially symmetric (mean $\Delta = -0.004$, $p = 0.908$), confirming that the choice of classifier does not systematically favour or penalise compositional representations.

For HyenaDNA and DNABERT-2, RF is significantly better than LP: RF wins in 37 and 43 of 57 datasets respectively, with mean differences of -0.036 ($p < 0.001$) and -0.041 ($p < 0.001$) in favour of RF. This indicates that the information encoded in these embeddings, although largely non-compositional, is not linearly decodable, consistent with our decomposition finding that their non-compositional residual is artefactual rather than structured. Caduceus shows the opposite pattern: LP outperforms RF in 40 of 57 datasets (mean $\Delta = +0.023$, $p = 0.002$), suggesting that its embedding geometry is more amenable to linear classification. Notably, this does not alter the main finding, even under LP, Caduceus does not systematically outperform k-mer features. Evo2 shows no significant difference between classifiers ($p = 0.121$).

These results support the use of RF as the primary classifier for two reasons. First, it does not systematically disadvantage any representation: for NTv3 and k-mers, the two methods that drive the main conclusions of the paper, the classifier choice is inconsequential. Second, RF avoids the assumption of linear separability, which is appropriate given the heterogeneous geometry of the embeddings tested. The significant LP advantage for Caduceus ($+0.023$ MCC) suggests that its embeddings are more linearly structured than those of other models, a finding consistent with its neutral decomposition profile (Section 4.3) and worth investigating in future work.

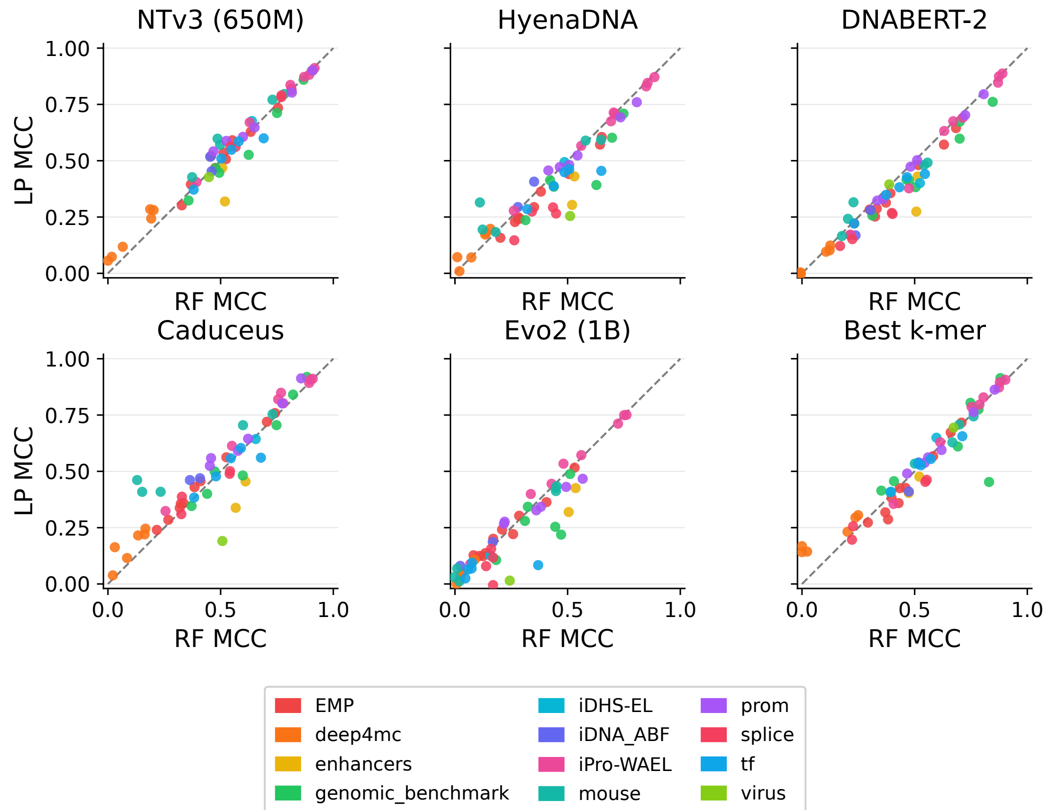

Figure 6: Linear probe (LP) MCC vs. Random Forest (RF) MCC across 57 classification datasets for five foundation models and the best k-mer baseline, coloured by biological category. Points above the diagonal favour LP; points below favour RF. NTv3 and k-mers show no systematic advantage for either classifier. HyenaDNA and DNABERT-2 are significantly better under RF, while Caduceus benefits from LP.

Table 2: Random Forest vs. linear probe comparison across 57 datasets. $\Delta = \text{LP} - \text{RF}$ MCC. Positive values favour LP; negative values favour RF. p-values from two-sided Wilcoxon signed-rank test.

Model	LP > RF	RF > LP	Mean Δ	Median Δ	p-value
NTv3-650M	25	14	$+0.009$	$+0.006$	0.094
HyenaDNA	10	37	-0.036	-0.032	$< 0.001^{***}$
DNABERT-2	4	43	-0.041	-0.036	$< 0.001^{***}$
Caduceus	40	12	$+0.023$	$+0.023$	0.002^{**}
Evo2-1B	21	26	-0.027	-0.006	0.121
Best k-mer	24	21	-0.004	$+0.004$	0.908

Appendix C K-mer Baseline Analyses

C.1 k-mer Selection by Biological Category

Figure 7 shows the distribution of MCC across $k \in \{4, 5, 6\}$ for each biological category. The optimal k varies systematically across categories: histone modification tasks (EMP) and coding-sequence benchmarks favour $k = 6$, while methylation tasks (iDNA_ABF) show a reverse trend with $k = 4$ performing best, consistent with the high GC content of CpG-dense regions. Promoter and enhancer tasks are largely insensitive to k selection. Splice-site tasks show high within-category variance at all k, reflecting the positional structure of splice signals that compositional features cannot fully capture.

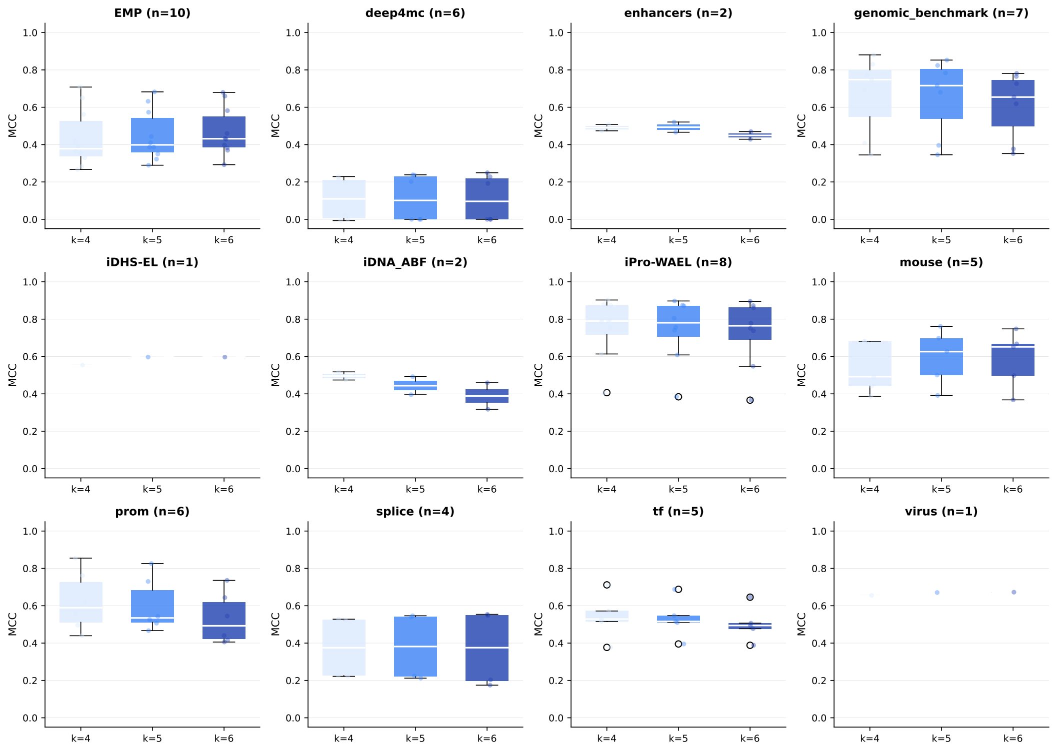

Figure 7: MCC distribution across $k \in \{4, 5, 6\}$ for each biological category, evaluated with a Random Forest classifier under 5-fold stratified cross-validation. Each box spans the interquartile range; the white bar indicates the median; circles denote outliers. The optimal k is strongly category-dependent and correlates with GC content (Spearman $r = -0.481$, $p < 0.001$): high-GC categories (methylation, promoters) favour smaller k, while low-GC categories (yeast histones) favour $k = 6$.

C.2 Canonical vs. Standard k-mer Features

DNA is double-stranded, meaning that a sequence and its reverse complement carry equivalent biological information. Canonical k-mers exploit this symmetry by collapsing each k-mer and its reverse complement into a single feature, reducing the feature dimension from 4^k to approximately $4^k/2$ for $k \geq 2$. We tested whether this normalisation improves classification performance relative to standard k-mers, which count each k-mer independently regardless of strand.

Figure 8 compares canonical and standard k-mer MCC across all 57 datasets for $k \in \{4, 5, 6\}$. Points lie almost uniformly on the diagonal, indicating negligible differences between the two representations.

For $k = 4$ and $k = 5$, the differences are not statistically significant (Wilcoxon signed-rank test, $p = 0.712$ and $p = 0.927$ respectively). For $k = 6$, canonical k-mers show a weakly significant improvement ($p = 0.037$), but the median effect remains small (Δmed $= +0.007$ MCC). Across all 171 comparisons, canonical k-mers improve over standard in 55.0% of cases, with a median difference of $+0.003$ MCC.

These results indicate that strand normalisation has no practical benefit for the classification tasks considered here. Although the $k = 6$ result reaches nominal significance, the effect size is smaller than the $\epsilon = 0.01$ threshold used throughout our evaluation and is unlikely to affect any dataset-level verdict. We therefore use standard k-mers as the baseline representation throughout the main analysis, which preserves the full 4^k-dimensional feature space and avoids the assumption that all biological signals are strand-symmetric, an assumption that may not hold for tasks with strand-specific regulatory elements such as promoters and splice sites.

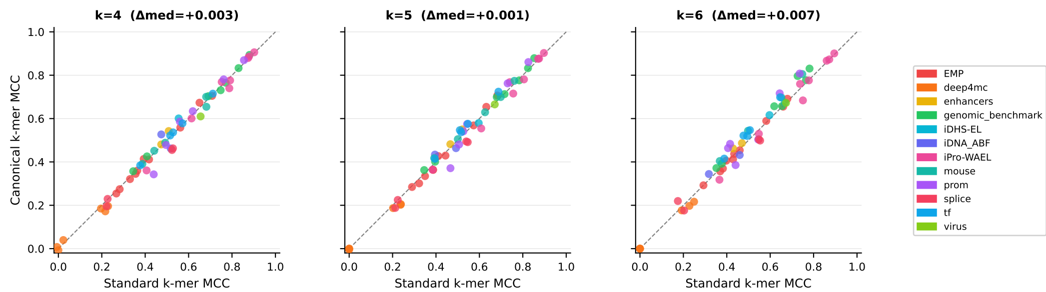

Figure 8: Canonical vs. standard k-mer MCC across 57 classification datasets for $k \in \{4, 5, 6\}$, coloured by biological category. The dashed diagonal indicates equal performance. Δmed denotes the median MCC difference (canonical − standard). Points cluster tightly around the diagonal at all values of k, indicating that strand normalisation has negligible effect on classification performance.

Appendix D Per-Dataset Classification Results

Table 3 reports MCC for all 57 classification datasets. The underlined value is the best k-mer baseline (selected across $k \in \{4, 5, 6\}$); **bold** indicates the overall best result per dataset across all methods. When the best k-mer is also the overall best, both styles apply.

Table 3: Per-dataset MCC across 57 classification benchmarks. Underline: best k-mer baseline. **Bold**: overall best per dataset. Category abbreviations: gb = genomic_benchmark, iPro = iPro-WAEL.

#	Dataset	Best k-mer	NTv3	HyenaDNA	DNABERT-2	Caduceus	Evo2
1	EMP/Yeast_H3	0.673	**0.734**	0.653	0.443	0.572	0.516
2	EMP/Yeast_H3K14ac	0.383	0.536	**0.544**	0.352	0.288	0.124
3	EMP/Yeast_H3K36me3	0.426	0.591	**0.594**	0.419	0.356	0.137
4	EMP/Yeast_H4	0.716	0.812	**0.827**	0.717	0.645	0.363
5	EMP/Yeast_H3K79me3	0.567	0.628	**0.656**	0.550	0.481	0.221
6	EMP/Yeast_H3K9ac	0.425	0.507	**0.538**	0.410	0.314	0.303
7	EMP/Yeast_H3K4me2	0.273	0.302	**0.348**	0.261	0.222	0.241
8	EMP/Yeast_H4ac	0.359	0.468	**0.495**	0.332	0.252	0.201
9	EMP/Yeast_H3K4me1	0.318	**0.521**	0.505	0.321	0.283	0.123
10	EMP/Yeast_H3K4me3	0.287	0.395	**0.400**	0.220	0.170	0.128
11	deep4mc/E.coli	0.141	0.056	**0.181**	−0.007	0.000	0.000
12	deep4mc/G.pickeringii	0.144	0.118	**0.148**	0.012	0.004	0.022
13	deep4mc/G.subterraneus	**0.168**	0.073	0.103	−0.003	−0.004	0.038
14	deep4mc/C.elegans	0.305	0.285	**0.353**	0.119	0.188	0.108
15	deep4mc/D.melanogaster	0.295	0.281	**0.357**	0.135	0.246	0.052
16	deep4mc/A.thaliana	0.232	0.243	**0.306**	0.106	0.224	0.046
17	enhancers/enhancer	**0.476**	0.468	0.425	0.446	0.430	0.426
18	enhancers/strength	**0.404**	0.319	0.344	0.254	0.274	0.319
19	gb/enhancer_cohn	0.456	0.467	**0.499**	0.456	0.379	0.343
20	gb/coding	0.804	0.859	**0.880**	0.742	0.673	0.488
21	gb/promoter_notata	**0.775**	0.712	0.744	0.634	0.598	0.427
22	gb/human_vs_worm	0.914	0.899	**0.925**	0.845	0.761	0.280
23	gb/open_chromatin	**0.414**	0.323	0.398	0.306	0.231	0.106
24	gb/enhancer_ensembl	**0.611**	0.526	0.558	0.445	0.519	0.219
25	gb/regulatory_region	0.453	0.446	**0.516**	0.476	0.334	0.254
26	iDHS-EL/DNase_I	0.649	**0.676**	0.665	0.581	0.401	0.126
27	iDNA_ABF/5mC	0.410	0.453	**0.522**	0.188	0.339	0.080
28	iDNA_ABF/6mA	0.539	0.517	**0.583**	0.298	0.450	0.186
29	iPro/Arab._TATA	0.796	0.820	**0.840**	0.747	0.632	0.534

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#	Dataset	Best k-mer	NTv3	HyenaDNA	DNABERT-2	Caduceus	Evo2
30	iPro/Arab._NonTATA	<u>0.829</u>	0.872	**0.885**	0.767	0.675	0.445
31	iPro/B.amyloliq.	<u>0.629</u>	**0.670**	0.651	0.390	0.526	0.399
32	iPro/HUVEC	<u>0.906</u>	**0.911**	0.907	0.901	0.887	0.751
33	iPro/GM12878	<u>0.872</u>	**0.882**	0.876	0.879	0.847	0.712
34	iPro/NHEK	<u>0.786</u>	**0.837**	0.761	0.729	0.691	0.572
35	iPro/R.capsulatus	<u>0.355</u>	**0.405**	0.334	0.150	0.332	0.088
36	iPro/Hela-S3	<u>0.895</u>	**0.901**	0.891	0.884	0.874	0.749
37	mouse/TFBS_1	<u>0.534</u>	**0.571**	0.364	0.249	0.334	0.031
38	mouse/TFBS_2	<u>0.708</u>	**0.796**	0.698	0.545	0.563	0.412
39	mouse/TFBS_3	<u>0.744</u>	**0.771**	0.683	0.565	0.491	0.434
40	mouse/TFBS_4	<u>**0.628**</u>	0.598	0.357	0.318	0.186	0.069
41	mouse/TFBS_5	<u>0.408</u>	**0.427**	0.372	0.197	0.217	0.012
42	prom/tata_300bps	<u>0.490</u>	**0.588**	0.547	0.354	0.323	0.267
43	prom/tata_70bps	<u>0.538</u>	0.542	0.491	0.340	**0.746**	0.276
44	prom/all_70bps	<u>0.562</u>	**0.606**	0.558	0.495	0.558	0.327
45	prom/all_300bps	<u>0.763</u>	**0.802**	0.779	0.759	0.702	0.431
46	prom/notata_70bps	<u>0.595</u>	**0.647**	0.604	0.538	0.585	0.343
47	prom/notata_300bps	<u>0.863</u>	**0.901**	0.868	0.855	0.795	0.467
48	splice/acceptors	<u>0.453</u>	**0.790**	0.414	0.392	0.274	0.156
49	splice/donors	<u>0.462</u>	**0.782**	0.412	0.384	0.352	0.079
50	splice/type_NT	<u>0.197</u>	**0.563**	0.401	0.186	0.317	0.117
51	splice/type_DNABERT	<u>0.256</u>	**0.561**	0.273	0.133	0.273	−0.005
52	tf/TFBS_1	<u>0.555</u>	**0.586**	0.559	0.454	0.495	0.095
53	tf/TFBS_2	<u>0.527</u>	**0.548**	0.519	0.429	0.474	0.064
54	tf/TFBS_3	<u>**0.535**</u>	0.510	0.488	0.381	0.348	0.069
55	tf/TFBS_4	<u>**0.409**</u>	0.372	0.357	0.265	0.221	0.025
56	tf/TFBS_5	<u>**0.656**</u>	0.600	0.531	0.454	0.441	0.084
57	virus/covid_variants	<u>**0.694**</u>	0.427	0.252	0.393	0.097	0.015

Note. MCC values are averaged over 5-fold stratified cross-validation. Negative values indicate performance below chance, occurring on datasets with extreme class imbalance or very low signal-to-noise ratio. Horizontal rules separate biological categories: EMP (1–10), deep4mc (11–16), enhancers (17–18), genomic_benchmark (19–25), iDHS-EL (26), iDNA_ABF (27–28), iPro-WAEL (29–36), mouse (37–41), prom (42–47), splice (48–51), tf (52–56), virus (57).

Appendix E Computational Efficiency

Figure 9 plots mean classification MCC against computational cost (GFLOPS per 500 bp sequence) for all methods, providing a direct cost-benefit comparison across the full evaluation.

K-mer features ($k = 6$) require approximately 1.9×10^{-5} GFLOPS per sequence, a near-zero cost reflecting the analytical nature of frequency counting, which involves no floating-point matrix operations. In contrast, FM inference costs range from 3.67 GFLOPS for NTv3-650M to 26.93 GFLOPS for DNABERT-2, despite its smaller parameter count (117M), due to the quadratic attention cost induced by BPE tokenisation generating more tokens per sequence. HyenaDNA-160k (6.30 GFLOPS) and Caduceus-Ph (5.21 GFLOPS) benefit from sub-quadratic architectures but remain approximately five orders of magnitude more expensive than k-mers.

The cost-benefit picture is stark: k-mers achieve mean MCC $= 0.535$ across 57 datasets at negligible computational cost, while the most expensive model (DNABERT-2, 26.93 GFLOPS) achieves mean MCC $= 0.427$, simultaneously $1.4 \times 10^6\times$ more costly and measurably less accurate. NTv3 is the only model that exceeds k-mer performance on average (mean MCC $= 0.562$), but does so at a cost of 3.67 GFLOPS per sequence, approximately $1.9 \times 10^5\times$ greater than k-mers. K-mer features therefore define the Pareto frontier of this evaluation: no foundation model simultaneously improves on k-mer accuracy and reduces computational cost.

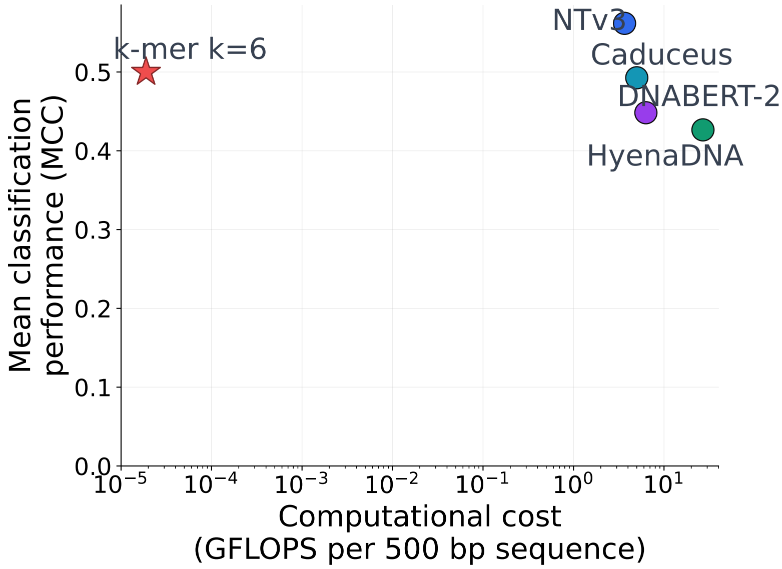

Figure 9: Mean classification MCC vs. computational cost (GFLOPS per 500 bp sequence, log scale) for k-mer features ($k = 6$, star) and five DNA foundation models (circles). K-mers occupy the Pareto frontier: they achieve competitive mean MCC (0.535) at a cost of $\sim 10^{-5}$ GFLOPS, approximately 10^5–$10^6\times$ lower than any FM. No foundation model simultaneously improves on k-mer performance and reduces computational cost.

FM embeddings justify their overhead only when the task requires non-compositional positional signals, most clearly splice-site classification, where NTv3 gains up to $+0.330$ MCC over k-mers, and when the approximately $10^5\times$ computational penalty is acceptable given the expected gain. For the majority of tasks evaluated here, including methylation, TFBS, and enhancer prediction, k-mers match or exceed FM performance at a cost that renders FM inference practically unnecessary.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction clearly state the three contributions: establishing k-mer frequency vectors as a rigorous baseline, introducing a geometric decomposition of FM embeddings, and characterising when non-compositional information is beneficial. All claims are explicitly scoped to the frozen-embedding regime (Section 1, paragraph following the three contributions) and are supported by experimental results across 57 classification datasets and two regression tasks reported in Section 4.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5.4 discusses six limitations: the frozen-embedding protocol and its deliberate scope, the use of a non-linear classifier with a linear decomposition, the linearity of the Ridge-based decomposition, the incomplete Evo2 decomposition results, potential selection bias in the dataset collection, and the absence of multi-species generalisation experiments.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper does not include formal theoretical results or proofs. The geometric decomposition in Section 3.4 is a methodological framework with clearly stated assumptions, but no theorems or lemmas are claimed.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 fully specifies the evaluation protocol, including dataset sources (Section 3.1), feature extraction procedures and model variants (Section 3.2), classifier hyperparameters (Random Forest, 500 trees, default scikit-learn settings) and cross-validation scheme (stratified 5-fold, Section 3.3), and the geometric decomposition procedure (Section 3.4). All five foundation models are publicly available. Per-dataset results are reported in full in Appendix B, Table 1.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: An anonymised repository containing all code and scripts necessary to reproduce the experiments is provided as supplemental material. All datasets used are publicly available through the sources cited in Section 3.1: the DNA Foundation Models Benchmark [13] and the Genomics Long-range Benchmark [29].

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3 specifies all details necessary to reproduce the experiments: dataset collections (Section 3.1), feature representations and model variants (Section 3.2), classifier hyperparameters and cross-validation scheme (Section 3.3), and the geometric decomposition procedure including the Ridge regression fitting protocol (Section 3.4). Computational costs are reported in Appendix B.3.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: MCC values are averaged over 5-fold stratified cross-validation. Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction across five simultaneous comparisons, and linear mixed-effects models with random intercepts per dataset, are reported in Section 4.1 and detailed in Appendix C, Table 2, for all five FM vs. best k-mer comparisons across 57 classification datasets.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Computational cost per sequence is reported in terms of GFLOPS in Appendix B.3 and Figure 8. All experiments were run on a single NVIDIA A100 GPU (80 GB HBM2e), with a total wall-clock runtime of approximately 20 GPU-hours covering FM embedding extraction across all 57 classification datasets and two regression tasks, k-mer feature computation, Random Forest training and evaluation, and the geometric decomposition.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: This work involves computational analysis of publicly available genomic sequence data and does not raise ethical concerns. No human subjects, sensitive personal data, or dual-use risks are involved.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: This work is foundational methodology research on genomic sequence representations. It does not introduce new models, datasets, or applications with direct societal impact. The findings may inform the design of future genomic foundation models, but no direct path to harmful applications is foreseen.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

    Answer: [N/A]

    Justification: The paper does not release new models, datasets, or pre-trained weights. All foundation models evaluated are existing publicly available systems. No safeguards beyond standard research practice are required.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All five foundation models and both dataset collections are cited with their original publications. Licenses are as follows: NTv3-650M (InstaDeep custom non-commercial licence, `https://huggingface.co/InstaDeepAI/NTv3_650M_pre`); HyenaDNA (BSD-3-Clause, `https:`

//huggingface.co/LongSafari/hyenadna-large-1m-seqlen-hf); DNABERT-2 (Apache 2.0, https://huggingface.co/zhihan1996/DNABERT-2-117M); Caduceus-Ph (Apache 2.0, https://huggingface.co/kuleshov-group/caduceus-ph_seqlen-131k_d_model-256_n_layer-16); Evo2-1B (Apache 2.0, https://huggingface.co/arcinstitute/evo2_1b_base). All models used in this work are for non-commercial academic research purposes, consistent with the terms of all licenses listed above. Dataset licenses are available at the respective repositories cited in Section 3.1.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: The paper does not release new datasets, models, or software packages as primary contributions. The geometric decomposition framework is described fully in Section 3.4 and is reproducible from the methodological description and the anonymised code repository provided as supplemental material.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: This work does not involve crowdsourcing or human subjects. All data used are publicly available genomic sequence datasets.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects research was conducted. IRB approval is not applicable.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [Yes]

    Justification: LLMs were used in two capacities that fall outside purely mechanical writing assistance: (1) language editing and structuring of the manuscript text, and (2) debugging of specific portions of the experimental codebase (e.g., identifying errors in data loading and plotting routines). LLMs were not used for any component of the core scientific methodology, including experimental design, statistical analysis, or interpretation of results, and did not influence the scientific claims of the paper.